STAT 709 Mathematical Statistics Lecture 2 - September 14, 2026

Lecture 2: Measure spaces, measures, and random variables

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Disclaimer: These notes are used for STAT 709 "Mathematical Statistics" at the University of Wisconsin–Madison. They are a learning aid rather than a substitute for a textbook, and corrections are welcome.

Do we need measure theory for every calculation in applied statistics or machine learning? Often the familiar rules for sums, densities, and expectations are enough. It is still useful to understand where that picture breaks down: a distribution may live on a curve, a statistical claim may involve infinitely many observations, or the data may themselves define a random measure.

Measure theory supplies a common language for these examples and for the limiting operations used throughout statistics. We will start with the models and the Riemann integral we already know, then identify the sets, probabilities, and functions needed to make those calculations precise. The goal is to explain what the abstraction buys us before developing it.

Recap from earlier lectures

From Lecture 1: sets, countability, eigenvalues

Reminder only: this block introduces no new exam requirements.

- **Countability**. A countable set is finite or can be listed in a sequence. Used in Section 2.
- **Set operations**. Complements, unions and intersections need not preserve the same properties for finite and infinite collections. Used in Section 3.
- **Symmetric matrices**. A real symmetric matrix has real eigenvalues and an orthonormal eigenbasis. Used in the optional random-spectrum example.

Learning goals

After this lecture, you should be able to:

- verify the defining properties of a σ-algebra and a measure;
- work with generated σ-algebras and the Borel σ-algebra;
- derive monotonicity, countable subadditivity, and continuity of measures;
- recognize Dirac, counting, Lebesgue, discrete, and probability measures;
- check measurability of a real-valued function using inverse images of half-lines;
- interpret a random variable as a measurable map and its distribution as a pushforward measure; and
- explain why densities and finite-dimensional sample spaces do not cover every probability model.

Priority guide

High	Core concepts to master for this lecture; will be directly examined.
Mid	Supporting concepts useful in other lectures; in scope of examinations.
Low	Enrichment supporting understanding; will not be examined.
Further reading	Optional self-study, references, extensions, and historical notes.

1 Why probability needs measure theory

1.1 Statistical motivations

High priority *Why we need measures to describe probability*

Two familiar probability models show why a list of outcomes and a density formula are not always sufficient.

An infinite sequence of observations. Suppose $X_1, X_2, \ldots$ are independent Bernoulli(1/2) observations. A finite sample lives in $\{0,1\}^n$, but an entire sequence lives in $\Omega = \{0,1\}^{\mathbb{N}}$. Asymptotic statistics asks about events such as

$$\left\{ \liminf_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} X_i > \frac{1}{3} \right\}.$$

How can we assign a probability to a condition involving infinitely many observations? We need one probability space for the whole sequence, with events closed under countable unions and intersections.

Specifying the first m tosses gives an event with probability 2^{-m}. The displayed limiting event cannot be decided from any fixed finite prefix; it is built using countably many unions and intersections of finite-sample events. Constructing that infinite product space is a later, optional topic; the need for closure under countable operations is already apparent. The countable-union/intersection representation is given in Exercise 9, with a complete construction in Section 6.

A distribution on a circle. Let $U \sim \mathrm{Unif}[0, 2\pi)$ and $(X, Y) = (\cos U, \sin U)$. All probability lies on the unit circle $S \subset \mathbb{R}^2$. Does this model have either a probability mass function or an ordinary two-dimensional density? The law exists, but it is neither discrete nor described by an area density.

In particular, there is no nonnegative Borel measurable function $f_{X,Y}$ such that, for every $a_1 < a_2$ and $b_1 < b_2$,

$$\int_{a_1}^{a_2} \int_{b_1}^{b_2} f_{X,Y}(x, y)\, dy\, dx = P\big(X \in [a_1, a_2],\ Y \in [b_1, b_2]\big). \tag{1}$$

Thus the problem is not merely that a continuous density is unavailable: even allowing a measurable density does not solve it. The two steps behind this claim—why the circle has area zero and why agreement on rectangles would determine its whole law—are worked out in the solution to Exercise 6.

Every individual point has probability zero, so a discrete mass function cannot describe the law. The circle also has area zero. Any density with respect to area would therefore give $P((X, Y) \in S) = 0$, whereas the model gives probability one. Nevertheless, probabilities are easy to interpret: an arc subtending an angle $\alpha \in [0, 2\pi]$ has probability $\alpha/(2\pi)$. The same exercise verifies the point and arc probabilities, including $\alpha = 0$ and $\alpha = 2\pi$.

Length, number of points, and probability are all instances of a *measure*. The central design is

sample space Ω	+	observable events $\mathcal{F}$	+	probability measure P.

A random variable maps this possibly complicated space into a familiar space such as $\mathbb{R}^d$ while preserving observable events. The aim is a collection of sets and functions rich enough for statistical models and stable under the countable operations used in limits.

Low priority *Motivating preview: a random measure—not examined*

Sometimes the observed object is itself a probability measure. Returning to the eigenvalues from Lecture 1, let X_n be a real symmetric $n \times n$ matrix with independent entries on and above the diagonal:

$$(X_n)_{ij} \sim N(0,1)\ (i < j), \qquad (X_n)_{ii} \sim N(0,2), \qquad (X_n)_{ji} = (X_n)_{ij}.$$

This is the Gaussian orthogonal ensemble (GOE). For $Y_n = X_n/\sqrt{n}$, its *empirical spectral distribution* assigns mass $1/n$ to each eigenvalue:

$$\mu_{Y_n} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i(Y_n)}, \qquad \mu_{Y_n}(B) = \frac{1}{n}\left|\{i : \lambda_i(Y_n) \in B\}\right|.$$

Here $\delta_a(B) = \mathbf{1}\{a \in B\}$ is a point mass. Thus $\mu_{Y_n}(B)$ is the fraction of eigenvalues in B; a histogram displays these masses after grouping them into bins. The measure is random because the matrix, and hence its eigenvalues, are random.

For this normalization, Wigner's semicircle law gives a deterministic limiting probability measure with density

$$\rho_{\mathrm{SC}}(x) = \begin{cases} \dfrac{1}{2\pi}\sqrt{4-x^2}, & |x| \le 2, \\ 0, & |x| > 2. \end{cases}$$

The convergence can be read through averages: for every bounded continuous function g, $n^{-1}\sum_i g(\lambda_i(Y_n))$ converges in probability to $\int_{-2}^{2} g(x)\rho_{\mathrm{SC}}(x)\,dx$; see [5]. What does it mean for an entire measure to converge? This is the motivating question, rather than a theorem to prove here. Section 5 returns to random measures after the basic definitions are available. For the theorem and proof, use the semicircle-law reference in Section 6.2.

Mid priority *A mental picture of the definitions ahead*

Instructor's Note *Conceptual understanding*

The following is an informal mental picture, not a substitute for the definitions. Even before learning the technical details, the gist is to set up the right mathematical workspace for sets, probabilities, functions, random variables, and limits.

- **Sets in $\mathbb{R}^d$:** for $d \ge 1$, think of the hierarchy

  $$\{\text{open or closed sets}\} \subseteq \{\text{Borel sets}\} \subsetneq \{\text{all subsets of } \mathbb{R}^d\}.$$

  Borel sets include the familiar geometric sets and are closed under countable set operations.

- **Functions on $\mathbb{R}^d$:** for real-valued functions, the picture is

  $$\{\text{continuous functions}\} \subseteq \{\text{Borel measurable functions}\} \subseteq \{\text{all functions}\}.$$

  Measurability keeps useful structure while allowing discontinuities.

- **Random variables:** familiar discrete and density-based models fit inside the broader class of measurable maps $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The circle example shows why this broader viewpoint matters for random vectors. The original sample space need not itself be Euclidean or have a Borel σ-algebra.

- **Integrable functions:** on a fixed compact interval $[a, b]$ with length measure, we have

  $$\{\text{bounded piecewise-continuous functions}\} \subseteq \{\text{Lebesgue integrable functions}\}.$$

  Here piecewise continuity means continuity on finitely many pieces with finite one-sided limits. Integrability depends on the domain and the measure; measurability alone does not imply integrability.

The aim is a class rich enough for the models we use and stable under the countable operations we need. The precise definitions tell us when this mental picture is valid. The relevant justifications are linked below: the Borel-generator argument, the measurability argument, and the integration references.

1.2 Riemann integration and the need for measure theory

Mid priority *Area, approximation, and two different limit problems*

From rectangles to an integral. For $f \geq 0$ on $[a,b]$, the Riemann integral represents area under the graph. More generally, take a bounded $f : [a,b] \to \mathbb{R}$ and a partition $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_m = b\}$. A tagged sum is

$$S(f, \mathcal{P}, t) = \sum_{i=1}^{m} f(t_i)(x_i - x_{i-1}), \qquad t_i \in [x_{i-1}, x_i].$$

Each term is height times interval length. The mesh $|\mathcal{P}| = \max_i (x_i - x_{i-1})$ measures the width of the largest interval. Riemann integrability means that these sums approach one finite value as $|\mathcal{P}| \to 0$, independently of the partitions and tags.

The lower and upper sums bound every tagged sum:

$$L(f, \mathcal{P}) = \sum_{i=1}^{m} \left(\inf_{x \in [x_{i-1}, x_i]} f(x) \right) (x_i - x_{i-1}),$$
$$U(f, \mathcal{P}) = \sum_{i=1}^{m} \left(\sup_{x \in [x_{i-1}, x_i]} f(x) \right) (x_i - x_{i-1}).$$

Equivalently, f is Riemann integrable if for every $\epsilon > 0$ there is a partition with $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon$. In that case,

$$\sup_{\mathcal{P}} L(f, \mathcal{P}) = \inf_{\mathcal{P}} U(f, \mathcal{P}) = \int_a^b f(x)\, dx.$$

This makes the approximation precise, but taking limits of functions raises two separate questions. For the equivalence of the tagged-sum and upper/lower-sum definitions, see the Riemann-integrability reference.

First problem: the limit may leave the Riemann class. Enumerate $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \ldots\}$, set $A_n = \{q_1, \ldots, q_n\}$, and let $f_n = \mathbf{1}_{A_n}$. Then

$$\int_0^1 f_n(x)\, dx = 0 \quad \text{for every } n, \qquad f_n(x) \uparrow \mathbf{1}_{\mathbb{Q} \cap [0,1]}(x).$$

Each f_n is Riemann integrable, but its pointwise limit is not.

A finite set can be covered by intervals of arbitrarily small total length, so the upper sums of f_n can be made arbitrarily small; the lower sums are zero. In contrast, every nondegenerate interval contains both rational and irrational points. For the limit function every upper sum is 1 and every lower sum is 0. Thus $\lim_n \int f_n = 0$ exists, while the Riemann integral of $\lim_n f_n$ does not exist. Lebesgue integration repairs this particular problem: the countable rational set has measure zero, so its indicator has Lebesgue integral zero as well. Exercise 10 asks for the details of this argument and the countable-null-set calculation; its solution is in Section 6.

Second problem: mass can escape to infinity. On $\mathbb{R}$, let $f_n(x) = \mathbf{1}_{[n, n+1]}(x)$. For each fixed x, the block eventually moves past x, so $f_n(x) \to 0$, although its area stays one:

$$\lim_{n \to \infty} \int_{\mathbb{R}} f_n(x)\, dx = 1 \neq 0 = \int_{\mathbb{R}} \lim_{n \to \infty} f_n(x)\, dx.$$

Here all the integrals exist, whether understood as improper Riemann integrals or Lebesgue integrals. Changing the definition of the integral does not make this exchange valid. The same exercise also explains which monotone-convergence hypothesis fails and why there is no common integrable dominating function.

The connection to integration with respect to a measure. Riemann sums weight function values by lengths of intervals. Measure theory extends this idea to measurable sets: a nonnegative simple function $s = \sum_{j=1}^{k} c_j \mathbf{1}_{B_j}$, with disjoint measurable B_j, has

$$\int s \, d\nu = \sum_{j=1}^{k} c_j \nu(B_j).$$

Here $c_j \geq 0$, and the convention $0 \cdot \infty = 0$ makes a zero-height piece contribute zero even when its supporting set has infinite measure. Length gives Lebesgue integration; counting gives sums; probability gives expectation. More general integrals are built by approximation. On a compact interval, a bounded Riemann-integrable function is Lebesgue integrable and the two integrals agree. The extension includes more functions and supplies precise convergence theorems; it does not permit unrestricted limit exchanges. Lecture 3 will distinguish monotone convergence and dominated convergence from Fatou's inequality. Nonnegativity alone does not give an equality, as the escaping block demonstrates. The simple-function construction and the agreement of the two integrals are supported by the integration references.

Instructor's Note *An informal picture of the extension*

The high-level idea is to **broaden the space of Riemann-integrable functions**. We work with sets more flexible than intervals, but retain useful rules for measuring them and adding countably many contributions. We still want intervals and their familiar integrals! This gives a better workspace for limits, rather than a promise that every limit can pass through an integral. The two examples above explain why both the larger workspace and the right convergence hypotheses are needed.

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2 σ-algebras and measurable spaces

High priority *Observable events and the Borel sigma-algebra*

Definition 1 (σ-algebra)**.** *Let Ω be a set. A collection $\mathcal{F} \subseteq 2^{\Omega}$ is a* σ-algebra *if:*

1. $\Omega \in \mathcal{F}$*;*
2. *$A \in \mathcal{F}$ implies $A^c = \Omega \setminus A \in \mathcal{F}$; and*
3. *$A_1, A_2, \ldots \in \mathcal{F}$ implies $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.*

The pair $(\Omega, \mathcal{F})$ is a measurable space*, and the elements of $\mathcal{F}$ are called* measurable sets*. In probability, they are called* events*.*

The three axioms immediately imply several closure properties:

$$\emptyset \in \mathcal{F}, \qquad \bigcap_{n=1}^{\infty} A_n \in \mathcal{F}, \qquad A \setminus B \in \mathcal{F} \quad \text{for } A, B, A_n \in \mathcal{F}.$$

For example, countable intersections follow from De Morgan's identity

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c.$$

The word *countable* matters: a σ-algebra is not required to be closed under arbitrary uncountable unions. For practice with these closure rules, see Exercise 1 and its solution.

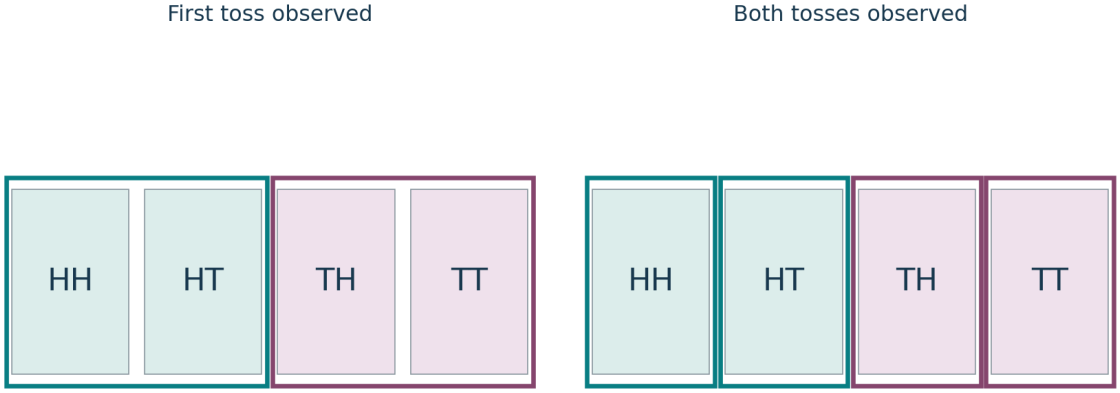
Example 1 (Information carried by a finite σ-algebra)**.** *For two coin tosses, let*

$$\Omega = \{HH, HT, TH, TT\}.$$

If we observe only the first toss, the available events form

$$\mathcal{F}_1 = \{\emptyset, \Omega, \{HH, HT\}, \{TH, TT\}\}.$$

This is a σ-algebra. The event $\{HH, TH\}$, which asks whether the second toss is heads, is deliberately absent: it cannot be determined from the available information.



Observing the second toss refines two information blocks into four singleton blocks.

Definition 2 (Generated σ-algebra)**.** *For a collection $\mathcal{C} \subseteq 2^\Omega$, the* σ-algebra generated by $\mathcal{C}$ *is*

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{A}\}.$$

It is the smallest σ-algebra containing $\mathcal{C}$.

The intersection in this definition is a σ-algebra, so the smallest one really exists. In Euclidean space, the *Borel σ-algebra* is

$$\mathcal{B}(\mathbb{R}^d) = \sigma\{O : O \subseteq \mathbb{R}^d \text{ is open}\}.$$

It contains all open and closed sets, intervals, rectangles, and every set obtained from these by countably many unions, intersections, and complements. Unless stated otherwise, random vectors taking values in $\mathbb{R}^d$ use $\mathcal{B}(\mathbb{R}^d)$ as their target σ-algebra. Exercise 11(1) proves that the intersection really is the smallest σ-algebra; see its solution.

Exercise prompt: generated events. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 3, 5\}$. Find $\sigma(\{A\})$. Is $\{1, 2\}$ measurable with respect to this σ-algebra? See Exercise 1(2–3) in Section 6 for the full statement and complete solution.

Remark 1 (Borel sets as a working collection)**.** *As an informal working rule, Borel sets are the* de facto *sets in Euclidean space that we use for observations and events. They include open and closed sets and remain Borel under countable unions, intersections, and complements, but they do not include every subset of $\mathbb{R}^d$. This is a useful middle ground between just intervals and all possible subsets.*[1]

[1] This practical rule does not mean that other measurable sets are never useful. Lebesgue measurable sets form a larger collection, obtained by completing Lebesgue measure; see Section 5. An optional cardinality comparison appears there as well.

Mid priority *Useful generators and limit operations on sets*

Remark 2 (Intervals and rectangles are Borel)**.** *Sets such as* $(a, b]$ *and* $[a, b]$ *are Borel. In* $\mathbb{R}^d$*, the rectangles*

$$(a_1, b_1] \times \cdots \times (a_d, b_d] = \prod_{j=1}^{d} (a_j, b_j]$$

are Borel as well, and these rectangles generate $\mathcal{B}(\mathbb{R}^d)$*.*

The same Borel σ*-algebra on* $\mathbb{R}$ *is generated by any of the following collections:*

$$\{(-\infty, a] : a \in \mathbb{R}\}, \qquad \{(a, b] : a < b\}, \qquad \{(a, b) : a < b\}.$$

It is enough to restrict the endpoints to rational numbers when open intervals are used. For instance,

$$(a, b) = \bigcup_{\substack{r,s \in \mathbb{Q} \\ a<r<s<b}} [r, s],$$

and the union is countable. These equivalent generators are useful because measurability needs to be checked only on a generating class. The countable-basis and rectangle steps are proved in Exercise 5; the generating-class test for functions is proved in Exercise 11.

Set limits are also written as

$$\liminf_{n\to\infty} A_n = \bigcup_{m=1}^{\infty} \bigcap_{n\geq m} A_n, \qquad \limsup_{n\to\infty} A_n = \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} A_n.$$

The first is the event that all but finitely many A_n occur; the second is the event that infinitely many A_n occur. Both are measurable whenever all A_n are measurable.

A useful limiting-set picture. If $a < b_n < b$ and $b_n \uparrow b$, then

$$(a, b) = \bigcup_{n=1}^{\infty} (a, b_n].$$

This is a concrete reason to want countable unions: the limiting interval is still measurable. The endpoint matters—if some $b_n = b$, the union instead contains b and equals $(a, b]$. The membership arguments for this identity and the two set limits appear in Exercise 9.

3 Measures and probability measures

High priority *Definition, calculations, and continuity of measures*

The length of $(a, b]$ is $b - a$, and the volume of a rectangle $\prod_{j=1}^{d} (a_j, b_j]$ is $\prod_{j=1}^{d} (b_j - a_j)$. A measure extends this familiar idea of size to sets in an abstract measurable space.

Definition 3 (Measure)**.** *Let* $(\Omega, \mathcal{F})$ *be a measurable space. A function* $\nu : \mathcal{F} \to [0, \infty]$ *is a* measure *if:*

1. $\nu(\emptyset) = 0$*; and*
2. *for every pairwise disjoint sequence* $A_1, A_2, \ldots \in \mathcal{F}$*,*

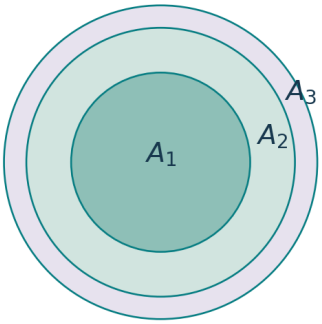
$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \nu(A_n).$$

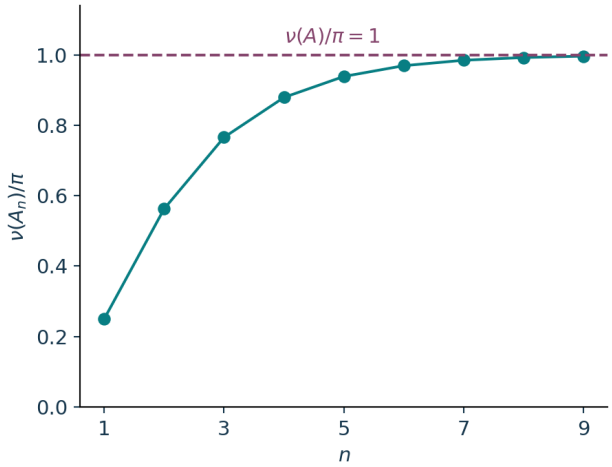
The triple $(\Omega, \mathcal{F}, \nu)$ *is a* measure space. *If* $\nu(\Omega) = 1$, *then* ν *is a* probability measure, *usually denoted by* P, *and* $(\Omega, \mathcal{F}, P)$ *is a* probability space.

Countable additivity yields the following rules that are used constantly.

Proposition 1 (Basic measure calculations)**.** *Let* ν *be a measure.*

1. *If* $A \subseteq B$, *then* $\nu(A) \le \nu(B)$.
2. *For arbitrary* $A_1, A_2, \ldots \in \mathcal{F}$,
$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) \le \sum_{n=1}^{\infty} \nu(A_n).$$
3. *If* $A_n \uparrow A$, *meaning* $A_1 \subseteq A_2 \subseteq \cdots$ *and* $A = \bigcup_n A_n$, *then*
$$\nu(A_n) \uparrow \nu(A).$$
4. *If* $A_n \downarrow A$ *and* $\nu(A_1) < \infty$, *then*
$$\nu(A_n) \downarrow \nu(A).$$





For disks A_n of radius $1 - 2^{-n}$, $\nu(A_n) = \pi(1 - 2^{-n})^2 \uparrow \pi$. The finite drawing illustrates an infinite sequence.

Proof. For monotonicity, write $B = A \cup (B \setminus A)$ as a disjoint union. For countable subadditivity, replace A_n by the disjoint sets
$$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{j<n} A_j, \quad n \ge 2.$$
Then $\bigcup_n B_n = \bigcup_n A_n$ and $\nu(B_n) \le \nu(A_n)$.

For continuity from below, use the disjoint increments $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$. Countable additivity gives
$$\nu(A) = \sum_{n=1}^{\infty} \nu(B_n) = \lim_{m\to\infty} \sum_{n=1}^{m} \nu(B_n) = \lim_{m\to\infty} \nu(A_m).$$
For continuity from above, apply continuity from below to $A_1 \setminus A_n \uparrow A_1 \setminus A$ and subtract from $\nu(A_1)$. The finiteness of $\nu(A_1)$ prevents the undefined subtraction $\infty - \infty$. □

Exercise prompt: an eventual head. For independent fair coin tosses, let A_n be the event of at least one head among the first n tosses. Find $P(A_n)$ and use continuity from below to find the probability of eventually seeing a head. See Exercise 2 in Section 6 and its complete solution.

Example 2 (Three fundamental measures)**.** *1. For $x \in \Omega$, the* Dirac measure *at x is*

$$\delta_x(A) = \mathbf{1}\{x \in A\}.$$

2. On a countable set with all subsets measurable, the counting measure *is $\nu_{\text{count}}(A) = |A|$, allowing the value ∞.*

3. The Lebesgue measure *m on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is the unique translation-invariant Borel measure normalized by*

$$m((a, b]) = b - a, \qquad -\infty < a < b < \infty.$$

Length as a probability. Restricting Lebesgue measure to $[0, 1]$ gives a probability measure, since $m([0, 1]) = 1$. Its measurable sets form the trace σ-algebra

$$\mathcal{B}_{[0,1]} = \{[0, 1] \cap B : B \in \mathcal{B}(\mathbb{R})\}.$$

This is the familiar uniform probability on $[0, 1]$: the probability of an interval inside $[0, 1]$ is its length. The trace collection inherits countable unions and complements relative to $[0, 1]$, and countable additivity is inherited from m.

Discrete probability measures are weighted sums of Dirac measures:

$$P = \sum_{k=1}^{\infty} p_k \delta_{x_k}, \qquad p_k \geq 0, \qquad \sum_{k=1}^{\infty} p_k = 1.$$

An absolutely continuous probability measure on $\mathbb{R}$ takes the familiar form $P(A) = \int_A f(x)\,dx$ for a nonnegative density f satisfying $\int f = 1$. Lecture 3 will unify both formulas as integration with respect to a measure. Exercise 3 verifies the discrete construction, including the interchange of its two nonnegative sums. The existence and uniqueness of Lebesgue measure and the density characterization are covered by the measure-construction references.

Mid priority *CDFs, sigma-finiteness, and finite product measures*

Cumulative distribution functions. For a probability measure P on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, define

$$F(x) = P((-\infty, x]).$$

Then F is nondecreasing and right-continuous, with $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to \infty$. Conversely, every function having these four properties determines a unique probability measure on $\mathcal{B}(\mathbb{R})$, with

$$P((a, b]) = F(b) - F(a).$$

Thus a CDF always exists, even when neither a probability mass function nor a Lebesgue density exists.

For a random vector with law P_X on $\mathbb{R}^d$, the same idea gives

$$F(x_1, \ldots, x_d) = P_X\big((-\infty, x_1] \times \cdots \times (-\infty, x_d]\big).$$

This is the multivariate cumulative *distribution* function. The forward properties of a CDF are proved in Exercise 12. For the converse existence and uniqueness assertion, see the cited construction and inverse-CDF proofs. The four one-dimensional conditions alone are not a multivariate characterization; rectangle increments must also be nonnegative.

Exercise prompt: right-continuity. For a probability measure P on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, let $F(x) = P((-\infty, x])$. If $x_n \downarrow x$, show that $F(x_n) \to F(x)$. Identify where finiteness of the measure is used. See Exercise 12 in Section 6 and its complete solution.

Sigma-finiteness. A measure ν is *σ-finite* if

$$\Omega = \bigcup_{n=1}^{\infty} E_n \quad \text{for measurable } E_n \text{ with } \nu(E_n) < \infty.$$

Probability measures, Lebesgue measure on $\mathbb{R}^d$, and counting measure on a countable set are σ-finite. This mild condition supports the product and Radon–Nikodym theorems used later. Concrete covers are the whole space for a probability measure, $[-n, n]^d$ for Lebesgue measure, and finite initial segments of an enumeration for a countable space.

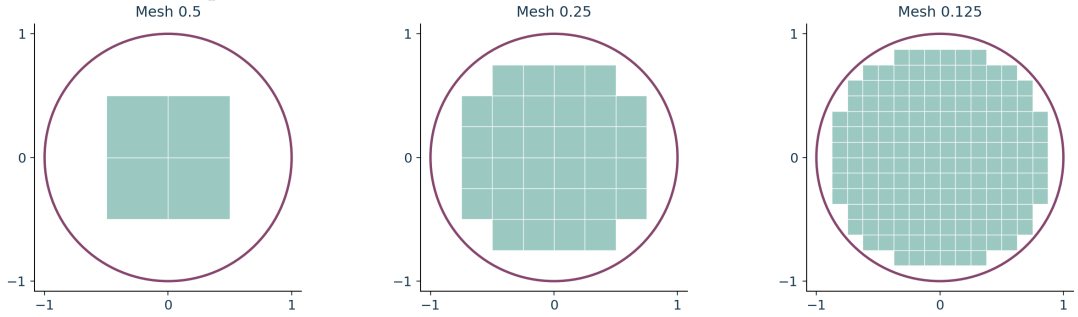
Product measures. For measurable spaces $(\Omega_i, \mathcal{F}_i)$, define

$$\mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma\{A_1 \times A_2 : A_i \in \mathcal{F}_i\}.$$

If ν_1 and ν_2 are σ-finite measures, there is a unique product measure $\nu_1 \otimes \nu_2$ satisfying

$$(\nu_1 \otimes \nu_2)(A_1 \times A_2) = \nu_1(A_1)\nu_2(A_2).$$

The theorem says much more than the rectangle formula: it extends that formula consistently to every event in the product σ-algebra. Lecture 3 will use the product measure in Tonelli's and Fubini's theorems. The general existence/uniqueness proof is in the product-measure reference; Exercise 7 works out the finite case and contrasts it with dependent coordinates.



Rectangles generate the product sigma-algebra; sets in that sigma-algebra need not themselves be rectangles. The squares approximate the open disk from inside.

Exercise prompt: independent Bernoulli coordinates. Let X_1, X_2 be independent Bernoulli(p) random variables, $0 \le p \le 1$. Find the four joint point probabilities and $P(X_1 + X_2 = 1)$. See Exercise 7 in Section 6 and its complete solution.

4 Measurable functions, random variables, and laws

High priority *Measurability tests and pushforward distributions*

The next step is to broaden our familiar class of continuous functions. The useful idea is to express a condition on a function in terms of the sets it pulls back. This is why inverse images are the central device.

For a map $f : \Omega \to \Lambda$ and a set $B \subseteq \Lambda$, the *inverse image* is

$$f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\}.$$

Inverse images preserve the set operations used by σ-algebras:

$$f^{-1}(B^c) = (f^{-1}(B))^c,$$
$$f^{-1}\left(\bigcup_n B_n\right) = \bigcup_n f^{-1}(B_n),$$
$$f^{-1}\left(\bigcap_n B_n\right) = \bigcap_n f^{-1}(B_n).$$

Definition 4 (Measurable function)**.** *Let $(\Omega, \mathcal{F})$ and $(\Lambda, \mathcal{G})$ be measurable spaces. A map $f : \Omega \to \Lambda$ is* $(\mathcal{F}, \mathcal{G})$-measurable *if*

$$f^{-1}(B) \in \mathcal{F} \qquad \text{for every } B \in \mathcal{G}.$$

It is enough to verify inverse images on a class that generates $\mathcal{G}$. In particular, a real-valued f is measurable if and only if

$$\{\omega : f(\omega) \le a\} \in \mathcal{F} \qquad \text{for every } a \in \mathbb{R}.$$

This is often the most useful calculation-based test. Exercise 11(2) proves the generating-class test; the solution explains why half-lines suffice.

Borel functions in practice. For maps between Euclidean spaces, a *Borel function* means a measurable function when both spaces carry their Borel σ-algebras. Unless specified otherwise, transformations used below are Borel measurable. A continuous function pulls open sets back to open sets, which explains why continuous functions are Borel measurable.

Proposition 2 (Operations that preserve measurability)**.**

1. *Every continuous $f : \mathbb{R}^d \to \mathbb{R}^k$ is Borel measurable.*
2. *If $f : (\Omega, \mathcal{F}) \to (\Lambda, \mathcal{G})$ and $g : (\Lambda, \mathcal{G}) \to (\Xi, \mathcal{H})$ are measurable, then $g \circ f$ is measurable.*
3. *If f and g are real-valued measurable functions, then $f+g$, fg, $\min(f, g)$, and $\max(f, g)$ are measurable.*
4. *If $f_1, f_2, \ldots$ are extended real-valued measurable functions, then $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$, and $\limsup_n f_n$ are measurable.*

These claims follow from inverse-image identities or composition with continuous maps. The worked supremum identity is left to the complete solution in Section 6. Complete proofs of the composition and arithmetic steps appear in Exercise 11; the countable supremum/infimum steps, including infinite values, appear in Exercise 4.

Exercise prompt: a measurable maximum. If $f_1, \ldots, f_n$ are real-valued measurable functions on $(\Omega, \mathcal{F})$, show that $M_n(\omega) = \max_{1 \le i \le n} f_i(\omega)$ is measurable. See Exercise 4(1) in Section 6 and its complete solution.

Composition retains a familiar property of continuous functions; closure under countable suprema and pointwise limits is an additional advantage of measurable functions. The rational-point indicators in Section 1.2 illustrate that measurability survives even when Riemann integrability does not. They do not give an example of continuous functions losing continuity: the finite-point indicators already have discontinuities.

Definition 5 (Random variable and random vector)**.** *On a probability space* $(\Omega, \mathcal{F}, P)$*, a* random variable *is a measurable map*

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

A measurable map into $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *is a* random vector*.*

These maps turn a potentially complicated sample space into numerical observations. The abbreviation "r.v." means random variable; informally, the term is sometimes also used for random vectors.

Definition 6 (Law or distribution)**.** *The* law *of a random variable* X *is the probability measure*

$$P_X = P \circ X^{-1}, \qquad P_X(B) = P(X \in B), \quad B \in \mathcal{B}(\mathbb{R}).$$

It is also denoted by $\mathcal{L}(X)$ *or* μ_X*.*

The notation $P(X \in B)$ is shorthand for $P(\{\omega : X(\omega) \in B\})$. Measurability is exactly what guarantees that this inverse image is an event to which P may be applied. Why the law is a *probability measure*, and why $P_{h(X)} = P_X \circ h^{-1}$, are verified in Exercise 11(4).

A finite inverse-image picture. On $\Omega = \{a, b, c, d\}$ with all subsets measurable and each outcome of probability $1/4$, define the random variable X by

ω	a	b	c	d
$X(\omega)$	-1	0	0	2

Choose a closed interval $B = [\ell, r]$ with integer endpoints $-2 \le \ell \le r \le 3$. Which outcomes belong to $X^{-1}(B)$, and how much probability does P_X assign to B? Compare a singleton interval with an interval that contains none of the observed values.

The map X sends outcomes to numbers. An inverse image instead pulls a set of numbers back to outcomes. Add the probabilities of those outcomes to obtain $P_X(B) = P(X^{-1}(B))$; repeated values of X collect multiple masses. No inverse function is needed, and a nonempty interval can have an empty inverse image.

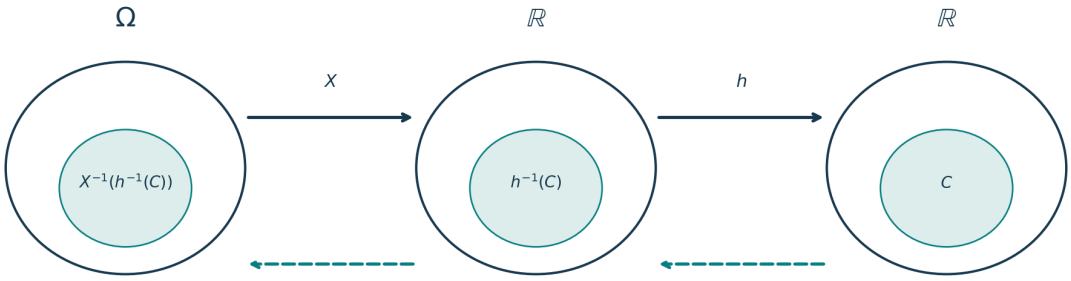
For $B = [0, 0] = \{0\}$, the inverse image is $\{b, c\}$ and $P_X(B) = 1/4 + 1/4 = 1/2$. For $B = [3, 3]$, it is empty and $P_X(B) = 0$. These are exact finite probabilities, not simulation estimates.

Example 3 (A law supported on a curve)**.** *For* $(X, Y) = (\cos U, \sin U)$*, the law* $P_{(X,Y)}$ *is the pushforward of the uniform law on* $[0, 2\pi)$*. If* $S = \{(x, y) : x^2 + y^2 = 1\}$*, then*

$$P_{(X,Y)}(S) = 1 \quad \textit{but} \quad m_2(S) = 0,$$

where m_2 *is two-dimensional Lebesgue measure. If a Lebesgue density existed, its integral over* S *would be zero, a contradiction. The law exists; only a particular representation of it as a two-dimensional density fails. For the area-zero calculation and the rectangle-identity argument, return to Exercise 6.*

Mid priority *How measurable maps encode statistical transformations*



Maps carry outcomes forward; inverse images pull events back into the original sample space.

If $X = (X_1, \ldots, X_d)$ is a random vector and $h : \mathbb{R}^d \to \mathbb{R}^k$ is Borel measurable, then $h(X)$ is a random vector. This single statement validates familiar constructions such as

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \max_{1\le i\le n} X_i, \qquad \|X\|_2, \qquad \mathbf{1}\{X \in B\}.$$

The law of $h(X)$ can be written as another pushforward:

$$P_{h(X)} = P_X \circ h^{-1}.$$

This viewpoint separates two objects that are often conflated:

- X is a function on the original sample space; and
- P_X is a probability measure on the observation space.

Most statistical calculations depend only on P_X, but almost-sure comparisons between several random variables depend on how they are jointly defined on Ω.

Remark 3 (An informal conclusion: why the abstraction helps)**.** *The point of the abstraction is to find a useful mathematical workspace in which our operations are rigorous. We want familiar sets such as intervals, closure under countable operations, and functions that preserve measurable events. Borel sets and Borel measurable functions are the standard starting point. This is not a blanket objection to using all subsets: on a finite or countable sample space that is natural. In Euclidean space, however, ordinary length cannot be extended to every subset while retaining both countable additivity and translation invariance. The Vitali-set reference gives the precise obstruction; its set-theoretic construction is optional.*

5 Technical extensions and further reading

Low priority *Completion and null-set modifications—not examined*

The Borel σ-algebra is generated by topology, but it is not complete with respect to Lebesgue measure: a subset of a Borel null set need not itself be Borel. The *Lebesgue σ-algebra* is the completion of $\mathcal{B}(\mathbb{R}^d)$ under Lebesgue measure. It contains every set of the form

$$B \cup N, \qquad B \in \mathcal{B}(\mathbb{R}^d), \quad N \subseteq Z, \quad Z \in \mathcal{B}(\mathbb{R}^d), \quad m(Z) = 0.$$

The completed measure assigns N measure zero.

Completion formalizes the principle that changing an object on a null set should not affect its integral or probability. It also explains why one must specify the relevant σ-algebra when using the phrase "measurable function": Borel measurability and Lebesgue measurability are closely related but not identical.

An informal picture of completion. Completion allows all changes to measurable sets on sets of measure zero. For real-valued functions on $\mathbb{R}^d$, keeping the target Borel σ-algebra fixed, this gives the hierarchy

$$\{\text{Borel measurable functions}\} \subseteq \{\text{Lebesgue measurable functions}\}.$$

Optional size comparison. For $d \geq 1$, the Borel sets have the same cardinality as $\mathbb{R}$, whereas the collection of all subsets of $\mathbb{R}^d$ has strictly larger cardinality:

$$|\mathcal{B}(\mathbb{R}^d)| = |\mathbb{R}| < |2^{\mathbb{R}^d}|.$$

Thus some subsets are not Borel. There are even non-Borel subsets of Borel null sets, which completion makes measurable. The set-theoretic argument is optional; see Tao's *An Introduction to Measure Theory*, Exercise 1.4.16. For the full completion construction, see the proof reference; Exercise 8 checks what happens to subsets of a null set. Return to the core Borel definition.

Low priority *Infinite products and random measures—not examined*

Finite product measures are not by themselves a construction of an infinite sequence $X_1, X_2, \ldots$. For finite-state or Euclidean coordinate spaces, the Kolmogorov extension theorem gives an infinite-dimensional probability measure when a consistent family of finite-dimensional distributions is supplied. This is the rigorous foundation for canonical sequence and stochastic-process models. The space assumptions and proof are stated in the extension-theorem reference.

A measure can itself also be data. Let Y_n be a symmetric random matrix, and write its eigenvalues as $\lambda_1, \ldots, \lambda_n$. Its empirical spectral distribution is the random probability measure

$$\mu_{Y_n} = \frac{1}{n} \sum_{i=1}^{n} \delta_{\lambda_i}.$$

For a normalized Gaussian orthogonal ensemble, this random measure converges under suitable conditions to the semicircle law with density

$$x \longmapsto \frac{1}{2\pi} \sqrt{4 - x^2}\, \mathbf{1}\{|x| \leq 2\}.$$

The density is interpreted as zero outside $[-2, 2]$, as in the piecewise formula in the introductory preview. This example motivates later notions of weak convergence for measures; see [5].

Return to the core definition of a law.

Further reading *References, construction theorems, and self-study*

For a probability-oriented development of measurable spaces, measures, and measurable maps, see Chapters 1–2 of [1]. Sections 1.1–1.2 of [4] emphasize the statistical viewpoint. A more analysis-centered treatment, including construction and completion of Lebesgue measure, appears in [2].

Useful self-study directions are:

- the Carathéodory extension theorem, which constructs a measure from its values on a simpler class of sets;
- the monotone-class and π–λ theorems, which reduce equality and measurability checks to generators;
- the Kolmogorov extension theorem for stochastic processes; and
- random probability measures and empirical distributions.

Key ideas

- A sigma-algebra specifies which sets can be assigned probabilities.
- Countable additivity yields monotonicity, subadditivity and continuity of measures.
- Random variables are measurable maps; their laws live on the value space.
- One measure-theoretic language includes discrete, continuous and singular models.

Looking ahead

Lecture 3 builds integrals from the measures introduced here. It then asks when a limit or a change in integration order preserves the answer. Continue to Lecture 3.

6 Exercises and complete solutions

In the future HTML version, this section should be rendered as an expandable block. In the PDF, all contents are shown.

The solution links in the main text lead directly to the relevant analysis below. Exercises 9–12 give additional justifications; Section 6.2 provides precise references for longer construction theorems.

High priority *Core practice: sigma-algebra calculations*

Exercise 1 (Closure properties and observed information). Let $\mathcal{F}$ be a σ-algebra on Ω.

1. Prove that $\mathcal{F}$ is closed under countable intersections and set differences.
2. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 3, 5\}$. Find $\sigma(A)$, the smallest σ-algebra containing A.
3. Is $\{1, 2\}$ measurable with respect to $\sigma(A)$? Interpret the answer in terms of information.

Solution. For $A_n \in \mathcal{F}$, each A_n^c belongs to $\mathcal{F}$, and hence

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c \in \mathcal{F}.$$

For $A, B \in \mathcal{F}$, $A \setminus B = A \cap B^c \in \mathcal{F}$.

A σ-algebra containing A must also contain A^c, $\emptyset$, and Ω. The collection

$$\{\emptyset, \Omega, A, A^c\} = \{\emptyset, \Omega, \{1, 3, 5\}, \{2, 4, 6\}\}$$

is itself a σ-algebra, so it is $\sigma(A)$. The set $\{1, 2\}$ is not in this collection. Observing only whether the outcome lies in A distinguishes odd from even outcomes, but it does not reveal the exact outcome. □

High priority *Core practice: continuity of probability*

Exercise 2 (An eventual head). Let $X_1, X_2, \ldots$ be independent fair coin tosses, with $X_i = 1$ for heads. Define

$$A_n = \{\text{at least one head occurs among the first } n \text{ tosses}\}.$$

1. Show that $A_n \uparrow A$, where A is the event that a head occurs at some finite time.
2. Compute $P(A_n)$ and use continuity from below to find $P(A)$.

3. Find the probability that all tosses are tails.

Solution. If a head appears among the first n tosses, it also appears among the first $n+1$, so $A_n \subseteq A_{n+1}$. The union $\bigcup_n A_n$ is exactly the event that a head appears at some finite index; hence $A_n \uparrow A$. Independence gives

$$P(A_n) = 1 - P(X_1 = \cdots = X_n = 0) = 1 - 2^{-n}.$$

By continuity from below,

$$P(A) = \lim_{n\to\infty} P(A_n) = 1.$$

The event that all tosses are tails is A^c, so it has probability zero. A probability-zero event need not be the empty set. □

High priority *Core practice: constructing and using a measure*

Exercise 3 (A discrete law). Let $x_1, x_2, \ldots$ be distinct points of $\mathbb{R}$, and let $p_k \geq 0$ satisfy $\sum_{k=1}^{\infty} p_k = 1$. Define

$$\mu(A) = \sum_{k=1}^{\infty} p_k \mathbf{1}\{x_k \in A\}, \qquad A \in \mathcal{B}(\mathbb{R}).$$

1. Show that μ is a probability measure.
2. Express μ using Dirac measures.
3. Write its CDF.
4. Explain why the same additivity argument gives counting measure on a countable sample space.

Solution. Clearly $\mu(\emptyset) = 0$ and $\mu(\mathbb{R}) = \sum_k p_k = 1$. If $A_1, A_2, \ldots$ are pairwise disjoint, then for each fixed k the point x_k belongs to at most one A_j. Since all terms are nonnegative,

$$\begin{aligned}\mu\left(\bigcup_{j=1}^{\infty} A_j\right) &= \sum_{k=1}^{\infty} p_k \sum_{j=1}^{\infty} \mathbf{1}\{x_k \in A_j\} \\ &= \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} p_k \mathbf{1}\{x_k \in A_j\} = \sum_{j=1}^{\infty} \mu(A_j).\end{aligned}$$

Here the interchange needs no integration theorem: for numbers $a_{jk} \geq 0$, both iterated sums equal

$$\sup_{J,K} \sum_{j=1}^{J}\sum_{k=1}^{K} a_{jk}.$$

Indeed, every finite choice of indices is contained in such a finite rectangle, and nonnegativity makes all these partial sums increase. The same reasoning allows the value $+\infty$. Thus μ is countably additive and is a probability measure. It can be written as

$$\mu = \sum_{k=1}^{\infty} p_k \delta_{x_k},$$

and its CDF is

$$F(t) = \mu((-\infty, t]) = \sum_{k=1}^{\infty} p_k \mathbf{1}\{x_k \leq t\}.$$

Putting unit weight on every point instead gives counting measure; the additivity calculation is unchanged, although the total mass can now be infinite. A single point with weight one gives the Dirac measure. □

High priority *Core practice: measurability under operations*

Exercise 4 (Maximum and limit superior). Let $X_1, X_2, \dots$ be real-valued random variables.

1. Prove directly from inverse images that $M_n = \max_{1\le i\le n} X_i$ is a random variable.
2. Prove that $\limsup_{n\to\infty} X_n$ is an extended real-valued measurable function.

Solution. For every $a \in \mathbb{R}$,

$$\{M_n \le a\} = \bigcap_{i=1}^{n} \{X_i \le a\} \in \mathcal{F},$$

so the half-line criterion shows that M_n is measurable.

Write

$$\limsup_{n\to\infty} X_n = \inf_{m\ge 1} \sup_{n\ge m} X_n.$$

For a countable family of measurable functions,

$$\left\{ \sup_{n\ge m} X_n \le a \right\} = \bigcap_{n\ge m} \{X_n \le a\} \in \mathcal{F},$$

so each $\sup_{n\ge m} X_n$ is measurable. For an infimum the convenient identity uses a *strict* inequality:

$$\{\inf_j Y_j < a\} = \bigcup_j \{Y_j < a\}, \qquad \{\inf_j Y_j \le a\} = \bigcap_{k=1}^{\infty} \bigcup_j \{Y_j < a + 1/k\}.$$

Each $\{Y_j < b\}$ is measurable because it is the union of $\{Y_j \le r\}$ over rational $r < b$. Thus countable infima are measurable, and applying this to $Y_m = \sup_{n\ge m} X_n$ proves the limsup claim. The formula $\liminf_n X_n = \sup_m \inf_{n\ge m} X_n$ proves the corresponding liminf assertion. These identities hold for extended real values as well; for example, $\{Y = +\infty\} = \bigcap_{k\ge 1}\{Y > k\}$ is measurable. One cannot replace the strict inequality above by $\{\inf_j Y_j \le a\} = \bigcup_j \{Y_j \le a\}$: take $Y_j = a + 1/j$. □

Mid priority *Supporting practice: Borel generators*

Exercise 5 (Half-lines and rectangles generate the Borel sets). Let $\mathcal{C} = \{(-\infty, a] : a \in \mathbb{R}\}$. Show that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. Then prove that open sets in $\mathbb{R}^d$ are countable unions of rational open boxes, and that either bounded closed boxes or bounded half-open boxes generate $\mathcal{B}(\mathbb{R}^d)$.

Solution. Each $(-\infty, a]$ is closed, hence Borel, so $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\mathbb{R})$. Conversely,

$$(a, b] = (-\infty, b] \setminus (-\infty, a] \in \sigma(\mathcal{C}).$$

Also,

$$(a, b) = \bigcup_{\substack{q\in\mathbb{Q} \\ a<q<b}} (a, q],$$

which is a countable union of sets in $\sigma(\mathcal{C})$. Every open subset of $\mathbb{R}$ is a countable union of bounded open intervals with rational endpoints, so every open set belongs to $\sigma(\mathcal{C})$. Since open sets generate $\mathcal{B}(\mathbb{R})$, we have $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{C})$.

Here is the countable-basis step in any dimension. Given x in an open set O, an open ball around x is contained in O. Choose rational endpoints $r_j < x_j < s_j$ so close to x_j that the closed box $\prod_j [r_j, s_j]$ lies in that ball. Thus the rational open boxes contained in O cover O, as do the rational closed boxes contained in O. There are only countably many such boxes because each is specified by $2d$ rational numbers. Hence closed boxes generate every open set and therefore the Borel sets; the reverse inclusion holds because closed boxes are Borel.

Half-open boxes are Borel as well, since

$$\prod_{j=1}^{d} (a_j, b_j] = \bigcap_{m=1}^{\infty} \prod_{j=1}^{d} (a_j, b_j + 1/m).$$

Conversely, every open box $\prod_j (a_j, b_j)$ is the union, over all sufficiently large m, of the half-open boxes $\prod_j (a_j, b_j - 1/m]$. This proves the other generating claim. It also justifies restricting to rational endpoints in the open-interval generators on $\mathbb{R}$. □

Mid priority *Supporting practice: a singular distribution*

Exercise 6 (Uniform distribution on the circle). Let $U \sim \mathrm{Unif}[0, 2\pi)$ and $Z = (\cos U, \sin U)$.

1. Show that every point has probability zero and that the circle $S = \{z : \|z\|_2 = 1\}$ has two-dimensional Lebesgue measure zero. For the latter, cover the circle by N small squares.
2. Deduce that the law has no discrete mass-function representation and is not absolutely continuous with respect to area.
3. Prove that no nonnegative Borel measurable density can satisfy Equation (1) for every rectangle. Explain why agreement on rectangles would imply agreement on S.
4. Let $B_\alpha = \{(\cos t, \sin t) : 0 \le t \le \alpha\}$ for $0 \le \alpha \le 2\pi$. Compute $P_Z(B_\alpha)$.

Solution. Points and area. The map $t \mapsto (\cos t, \sin t)$ is one-to-one on $[0, 2\pi)$, so the inverse image of a point is empty or a singleton. Uniform probability assigns it zero mass. To check the area assertion, put $h = 2\pi/N$ and $t_k = kh$, $0 \le k < N$. On $[t_k, t_k + h]$, the mean value theorem gives $|\cos t - \cos t_k| \le h$ and $|\sin t - \sin t_k| \le h$. The arc is therefore contained in the closed square centered at $(\cos t_k, \sin t_k)$ with side $2h$. The N squares cover S, so

$$0 \le m_2(S) \le N(2h)^2 = \frac{16\pi^2}{N} \longrightarrow 0.$$

The set S is closed, hence Borel, so this proves $m_2(S) = 0$.

Consequences for the law. A discrete probability law must put total mass one on a countable set. Here every countable set has probability zero by countable additivity. Also $P_Z(S) = 1$ while $m_2(S) = 0$, so $P_Z \not\ll m_2$: absolute continuity would require every area-zero set to have probability zero.

Why the rectangle equation fails. Suppose a function $f \ge 0$ satisfied Equation (1). Define the measure $\nu_f(A) = \int_A f \, dm_2$ on Borel sets. We use two standard facts about Lebesgue integration: a nonnegative measurable function defines a measure this way, and an integral over a null set is zero. The iterated integral equals this area integral on rectangles by Tonelli's theorem; see the integration and product references. For $m \ge 1$, the assumed equation gives $\nu_f([-m, m]^2) = P_Z([-m, m]^2) = 1$; continuity from below yields $\nu_f(\mathbb{R}^2) = 1$.

Now apply the following uniqueness fact: *two finite measures with the same total mass that agree on a generating collection closed under finite intersections agree on the whole generated σ-algebra.* Such a collection is called a π-system. Its proof is the finite-measure uniqueness theorem cited below. Here use bounded closed rectangles (including degenerate ones), together with $\emptyset$ and $\mathbb{R}^2$. These generate the Borel sets by Exercise 5. For a degenerate rectangle, the area integral is zero and its intersection with S contains

at most two points, so its P_Z-mass is also zero. All the hypotheses are met. Consequently $\nu_f = P_Z$ on Borel sets, but

$$1 = P_Z(S) = \nu_f(S) = \int_S f \, dm_2 = 0,$$

a contradiction. This supplies the step from rectangles to the curved support that is implicit in the main-text explanation.

Arc probabilities. Except for endpoints, which have probability zero, the event $\{Z \in B_\alpha\}$ is $\{0 \le U \le \alpha\}$. Therefore

$$P_Z(B_\alpha) = \frac{\alpha}{2\pi}.$$

This is a one-dimensional uniform law along the circle, not a two-dimensional density. At $\alpha = 0$ the arc is one point and the probability is zero; at $\alpha = 2\pi$ it is the whole circle and the probability is one. □

Mid priority *Supporting practice: finite product measures*

Exercise 7 (Two independent Bernoulli variables). Let ν be the Bernoulli(p) measure on $\{0, 1\}$, where $0 \le p \le 1$, so $\nu(\{1\}) = p$. On $\{0, 1\}^2$ use the product measure $\nu \otimes \nu$. Let $X_i(x_1, x_2) = x_i$ be the coordinate maps.

1. Compute the probability of every singleton (x_1, x_2).
2. Compute $P(X_1 + X_2 = 1)$.
3. Explain where the product assumption enters.

Solution. For $x_1, x_2 \in \{0, 1\}$, the rectangle formula gives

$$P((X_1, X_2) = (x_1, x_2)) = p^{x_1+x_2}(1-p)^{2-x_1-x_2}.$$

Equivalently, the four masses are $(1-p)^2$, $p(1-p)$, $p(1-p)$, and p^2 at $(0,0), (0,1), (1,0), (1,1)$, respectively. This list also covers $p = 0$ and $p = 1$ without needing to interpret 0^0 in the compact formula. Hence

$$P(X_1 + X_2 = 1) = P(1,0) + P(0,1) = 2p(1-p).$$

The multiplication of marginal probabilities is exactly the defining rectangle property of the product measure. A general joint measure with the same Bernoulli marginals need not assign these probabilities and need not make X_1 and X_2 independent. For example, setting $X_1 = X_2 = W$ for one Bernoulli(p) variable gives the same two marginals but $P(X_1 + X_2 = 1) = 0$. For $0 < p < 1$ this differs from the independent answer. □

Low priority *Optional practice: completion—not examined*

Exercise 8 (Subsets of null sets). Let $(\Omega, \mathcal{F}, \nu)$ be a measure space and let $N \in \mathcal{F}$ satisfy $\nu(N) = 0$. Define the completion

$$\overline{\mathcal{F}} = \{A \cup M : A \in \mathcal{F},\ M \subseteq N_A \text{ for some } N_A \in \mathcal{F} \text{ with } \nu(N_A) = 0\}.$$

Explain why every subset $M \subseteq N$ belongs to $\overline{\mathcal{F}}$ and why its completed measure must be zero.

Solution. Take $A = \emptyset$ and $N_A = N$. Then $M = \emptyset \cup M$ has the required representation, so $M \in \overline{\mathcal{F}}$. Any extension that remains monotone must satisfy

$$0 \le \overline{\nu}(M) \le \overline{\nu}(N) = \nu(N) = 0.$$

Thus $\overline{\nu}(M) = 0$. The completion adds all such subsets while leaving the measure of every original event unchanged. The separate tasks of proving that $\overline{\mathcal{F}}$ is a σ-algebra and that the extension is well-defined are supplied by the completion theorem and its proof. □

6.1 Further justification exercises

High priority *Core practice: countable descriptions of limiting events*

Exercise 9 (From finite observations to a limiting event).

1. Explain why $\liminf_n A_n$ consists of points in all sufficiently late A_n, whereas $\limsup_n A_n$ consists of points in infinitely many A_n. If $a < b_n < b$ and $b_n \uparrow b$, prove $\bigcup_n (a, b_n] = (a, b)$.

2. On $\Omega = \{0,1\}^{\mathbb{N}}$, let $X_i(\omega) = \omega_i$ and $S_n = n^{-1} \sum_{i=1}^{n} X_i$. Use the σ-algebra generated by events specifying a finite initial segment. Show that

$$\{\liminf_n S_n > 1/3\} = \bigcup_{\substack{r \in \mathbb{Q} \\ r > 1/3}} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{S_n > r\},$$

   and deduce measurability.

3. Explain why no fixed finite initial segment determines membership in this event.

Solution. **Set limits.** Membership in $\bigcup_N \bigcap_{n \geq N} A_n$ means that some N works for every later n; membership in $\bigcap_N \bigcup_{n \geq N} A_n$ means that every N has a later successful n. These are exactly "eventually always" and "infinitely often." For the intervals, every point in the union lies strictly between a and b. Conversely, if $a < x < b$, convergence $b_n \to b$ implies $b_n > x$ for some n, so $x \in (a, b_n]$. If a nondecreasing sequence reaches b, the union instead includes b.

The limiting event. If $\liminf_n S_n > 1/3$, choose a rational r strictly between these two numbers. By the definition $\liminf_n S_n = \sup_N \inf_{n \geq N} S_n$, some tail has infimum greater than r, so all its S_n exceed r. Conversely, if eventually $S_n > r$ for some $r > 1/3$, then $\liminf_n S_n \geq r > 1/3$. Notice the weak inequality in this converse: a sequence strictly above r can converge to r.

For each n, the set $\{S_n > r\}$ is a finite union of events specifying the first n zeros and ones: include exactly the strings whose average exceeds r. It is therefore measurable. Both unions in the displayed identity and the intersection are countable, so the limiting event is measurable as well. This argument identifies an event; it does not yet calculate its probability. Existence of the infinite-sequence probability law is the separate construction cited in the infinite-product reference.

Why finite observations do not suffice. After any fixed initial segment, append zeros forever or append ones forever. In the first case $S_n \to 0$; in the second $S_n \to 1$. These two sequences agree on all the observed coordinates but give different answers to the limiting event. Countable set operations let us describe it without requiring a finite observation to decide it. □

Mid priority *Supporting practice: the two integration pitfalls*

Exercise 10 (What fails when a limit passes through an integral?).

1. For an enumeration $\{q_1, q_2, \ldots\}$ of $\mathbb{Q} \cap [0,1]$, set $f_n = \mathbf{1}_{\{q_1, \ldots, q_n\}}$. Prove the Riemann-integral assertions in Section 1.2, and compute the Lebesgue integral of the limit.

2. For $g_n = \mathbf{1}_{[n, n+1]}$ on $\mathbb{R}$, prove pointwise convergence to zero while $\int g_n \, dm = 1$. Explain why the sequence is not increasing and why no integrable function can dominate every g_n.

Solution. **Finite rational sets and their limit.** Every interval of positive length contains an irrational point, so every lower sum of f_n is zero. Given $\epsilon > 0$, enclose its finitely many nonzero points in small intervals of total length less than ϵ. Choose the endpoints away from those points and include them in a partition;

use one-sided intervals at 0 or 1. Only these small intervals can contribute to the upper sum, so that sum is less than ϵ. The Darboux criterion gives $\int_0^1 f_n\,dx = 0$.

Pointwise, $f_n \uparrow f = \mathbf{1}_{\mathbb{Q}\cap[0,1]}$. Every nondegenerate partition interval contains both rational and irrational points, so f has upper sum 1 and lower sum 0 for every partition. Thus f is not Riemann integrable. For Lebesgue measure, a singleton can be enclosed in intervals of arbitrarily small length and hence has measure zero. Countable subadditivity gives $m(\mathbb{Q}\cap[0,1]) = 0$. Therefore

$$\int_0^1 f\,dm = m(\mathbb{Q}\cap[0,1]) = 0 = \lim_n \int_0^1 f_n\,dm.$$

The limit has left the Riemann class, but the Lebesgue integrals do agree with the limit of the integrals in this example.

An escaping block. For any fixed x, eventually $n > x$, so $g_n(x) = 0$. Each interval $[n, n+1]$ has length one, giving

$$\int \lim_n g_n\,dm = 0 \quad \text{but} \quad \lim_n \int g_n\,dm = 1.$$

At $x = n + 1/2$, $g_n(x) = 1$ and $g_{n+1}(x) = 0$, so the sequence is not increasing. If an integrable nonnegative g dominated every g_n, it would satisfy $g \geq 1$ on $\bigcup_{n\geq 1}[n, n+1] = [1, \infty)$, which forces $\int g\,dm = \infty$, a contradiction. The same argument works for domination almost everywhere: the union of the countably many exceptional null sets is still null.

These are the missing hypotheses for the monotone and dominated convergence theorems, respectively, which will be developed in Lecture 3. Fatou's inequality would only assert $0 \leq 1$ here and causes no contradiction. Their proofs are also available in [1, Theorems 1.5.5, 1.5.7, and 1.5.8]. □

High priority *Core practice: justifying measurability tests*

Exercise 11 (Generators, operations, and distributions).

1. Prove that the intersection of all σ-algebras on Ω containing $\mathcal{C}$ is a σ-algebra and is the smallest one containing $\mathcal{C}$.

2. If $\mathcal{G} = \sigma(\mathcal{C})$, prove that $f : (\Omega, \mathcal{F}) \to (\Lambda, \mathcal{G})$ is measurable whenever $f^{-1}(C) \in \mathcal{F}$ for every $C \in \mathcal{C}$. Deduce the half-line test for a real-valued function.

3. Prove that continuous maps between Euclidean spaces are Borel measurable, that compositions of measurable maps are measurable, and that the arithmetic operations in Proposition 2 preserve measurability.

4. Prove that $P_X(B) = P(X^{-1}(B))$ is a probability measure and that $P_{h(X)}(B) = P_X(h^{-1}(B))$ for measurable h.

Solution. 1. The generated σ-algebra. The family being intersected is nonempty because 2^Ω belongs to it. Its intersection contains Ω and $\mathcal{C}$. If A belongs to every member of the family, so does A^c; if each A_n belongs to every member, so does $\bigcup_n A_n$. These observations prove the σ-algebra axioms. The intersection is contained in every σ-algebra containing $\mathcal{C}$, which proves minimality.

2. Checking generators. Define

$$\mathcal{D} = \{B \subseteq \Lambda : f^{-1}(B) \in \mathcal{F}\}.$$

Since inverse images preserve the whole space, complements, and countable unions, $\mathcal{D}$ is a σ-algebra on Λ. The hypothesis says $\mathcal{C} \subseteq \mathcal{D}$, so $\mathcal{G} = \sigma(\mathcal{C}) \subseteq \mathcal{D}$. This is exactly measurability of f. By Exercise 5, the half-lines $(-\infty, a]$ generate $\mathcal{B}(\mathbb{R})$; checking $\{f \leq a\} \in \mathcal{F}$ for every $a \in \mathbb{R}$ therefore suffices.

3. Continuous maps, compositions, and arithmetic. A continuous map has open inverse images of open sets. Open sets generate the Borel σ-algebra, so part 2 proves Borel measurability. For measurable f and h, the identity $(h \circ f)^{-1}(B) = f^{-1}(h^{-1}(B))$ proves composition is measurable.

If f and g are real-valued measurable functions on Ω, the map $F = (f, g)$ is measurable: the inverse image of a closed rectangle $[a, b] \times [c, d]$ is the intersection $\{a \le f \le b\} \cap \{c \le g \le d\} \in \mathcal{F}$, and these rectangles generate $\mathcal{B}(\mathbb{R}^2)$ by Exercise 5. The maps $(u, v) \mapsto u + v, uv, \min(u, v), \max(u, v)$ are continuous. Composing each with F proves all four arithmetic claims. The same rectangle argument proves that a finite vector is measurable exactly when each coordinate is measurable; the reverse direction uses the continuous coordinate projections. Countable extrema and limits are handled separately in Exercise 4.

4. Pushing a probability measure forward. The empty set has empty inverse image, and the entire target has inverse image Ω, so $P_X(\emptyset) = 0$ and $P_X(\Lambda) = 1$. For pairwise disjoint measurable B_j, their inverse images are measurable and disjoint. Consequently

$$P_X\Big(\bigcup_j B_j\Big) = P\Big(\bigcup_j X^{-1}(B_j)\Big) = \sum_j P(X^{-1}(B_j)) = \sum_j P_X(B_j).$$

Thus P_X is a probability measure. Finally,

$$P_{h(X)}(B) = P((h \circ X)^{-1}(B)) = P(X^{-1}(h^{-1}(B))) = P_X(h^{-1}(B)),$$

which proves the distribution formula used in the main text. □

Mid priority *Supporting practice: the properties of a CDF*

Exercise 12 (From a probability measure to its CDF). For a probability measure P on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, let $F(x) = P((-\infty, x])$. Prove monotonicity, right-continuity, the limits $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to \infty$, and

$$P((a, b]) = F(b) - F(a) \qquad (a < b).$$

Identify where finiteness of the measure is used. Does this argument also construct a probability measure from a candidate F?

Solution. If $x \le y$, then $(-\infty, x] \subseteq (-\infty, y]$, so monotonicity of P gives $F(x) \le F(y)$. If $x_n \downarrow x$, the corresponding half-lines decrease to $(-\infty, x]$. Continuity from above applies because their probabilities are finite and gives $F(x_n) \downarrow F(x)$. In particular $F(x + 1/n) \to F(x)$; monotonicity squeezes $F(y)$ between $F(x)$ and $F(x + 1/n)$ whenever $x < y \le x + 1/n$. This proves right-continuity for arbitrary approaches from the right.

The sets $(-\infty, n]$ increase to $\mathbb{R}$, while $(-\infty, -n]$ decrease to $\emptyset$. Continuity from below and above, respectively, give $F(n) \to 1$ and $F(-n) \to 0$. Monotonicity extends these integer limits to the limits along all real x. The decreasing-set steps use finiteness; the increasing-set step does not. Finally,

$$(-\infty, b] = (-\infty, a] \dot\cup (a, b]$$

is a disjoint union, so additivity gives the interval formula.

This proves the properties of the CDF of an existing measure. The converse is a separate existence theorem: every nondecreasing, right-continuous F with these two limiting values is the CDF of a probability measure. Its construction, and uniqueness from the half-line probabilities, are proved in the CDF and Stieltjes-measure references below. □

6.2 References for longer proofs

The results used in the main sections keep their assigned priority. The references below supply longer proofs and construction details; they do not add new examination requirements. Exercises above give the local arguments students can work through with this lecture's tools.

Uniqueness from rectangles. The π–λ theorem and measure uniqueness are proved in [1, Theorems A.1.4–A.1.5]; see the author's fifth-edition text. The finite-measure version stated in Exercise 6 is the fact needed for Equation (1).

Existence of measures and CDFs. Durrett's Theorem 1.1.4 constructs the unique Stieltjes measure with $\mu((a,b]) = F(b) - F(a)$; $F(x) = x$ yields Lebesgue measure on $\mathbb{R}$. Theorem 1.2.2 constructs a random variable with a prescribed CDF. Theorem A.1.5 supplies uniqueness, and Theorem A.4.8 is the Radon–Nikodym theorem giving a density for σ-finite $\nu \ll \mu$. These are proved in the same text.

Riemann and Lebesgue integration. In [3], Theorem 6.6 proves the upper/lower-sum criterion, Theorem 6.10 handles bounded functions with finitely many discontinuities, and Theorem 11.33 proves agreement of the Riemann and Lebesgue integrals and the almost-everywhere continuity criterion; see the third-edition text. The simple-function construction is developed in [1, Sections 1.4–1.5]. Lecture 3 will develop these integration tools systematically.

Products and density integrals. Durrett's Theorems 1.7.1–1.7.2 prove existence/uniqueness of σ-finite product measures and Fubini's theorem, including the nonnegative case used as Tonelli's theorem. Sections 1.4–1.5 justify the facts about $\nu_f(A) = \int_A f\, d\mu$ used in Exercise 6: integration over a null set gives zero directly from simple-function approximation; for disjoint A_j, monotone convergence applied to $f \sum_{j=1}^{n} \mathbf{1}_{A_j}$ gives countable additivity. Proofs are in the author's text.

Completion (optional). Durrett's Theorem A.2.3 proves that completion produces a σ-algebra and a well-defined measure; the following example in Section A.2 constructs a nonmeasurable set. Another full construction is Proposition 1.2.18 of Tao's *An Introduction to Measure Theory*. Exercises 1.4.15–1.4.16 there give a guided, optional route to the cardinality comparison in Section 5; these are textbook exercises, not supplied complete solutions.

Infinite products and random matrices (optional proofs). Theorem 2.4.3 of Tao's measure-theory text states and proves Kolmogorov's extension theorem with its space and regularity hypotheses; finite-state and Euclidean coordinate spaces cover the models here. For the GOE preview, see Theorem 1 and the moment-method proof in Section 2 of Tao's *254A, Notes 4: The semi-circular law*.

References

[1] Rick Durrett. *Probability: theory and examples*, volume 49. Cambridge university press, 5th edition, 2019.

[2] W. Rudin. *Real and Complex Analysis*. Higher Mathematics Series. McGraw-Hill Education, 1987.

[3] Walter Rudin. *Principles of mathematical analysis*. McGraw-Hill, New York, 3 edition, 1976.

[4] Jun Shao. *Mathematical Statistics*. Springer Science & Business Media, 2nd edition, 2013.

[5] Terence Tao. *Topics in random matrix theory*, volume 132. American Mathematical Soc., 2012.

What changed

October 14, 2026. Added issue-reporting links to the webpage and PDF. The mathematics and priority assignments are unchanged in this update.